Snowflake SnowPro Core Certification

Exam Preparation Series with Practice Questions 2025

Chapter 1

1.1 Outline Key Features Of The Snowflake AI Data Cloud.

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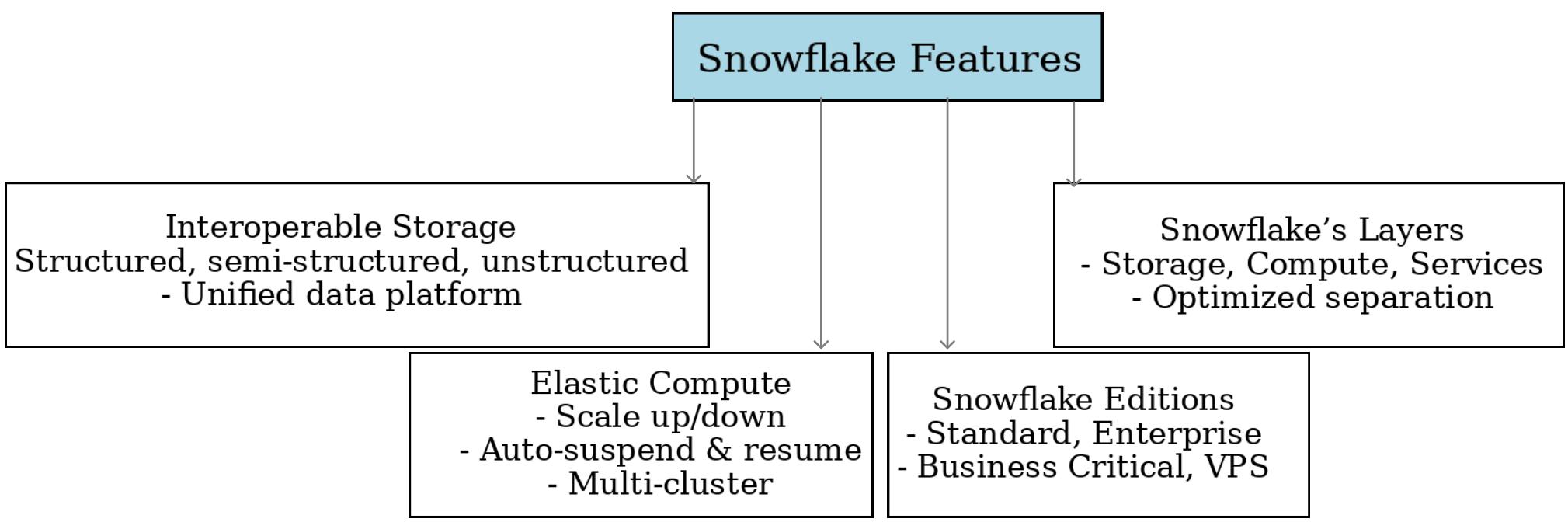
Outline key features of the Snowflake AI Data Cloud.

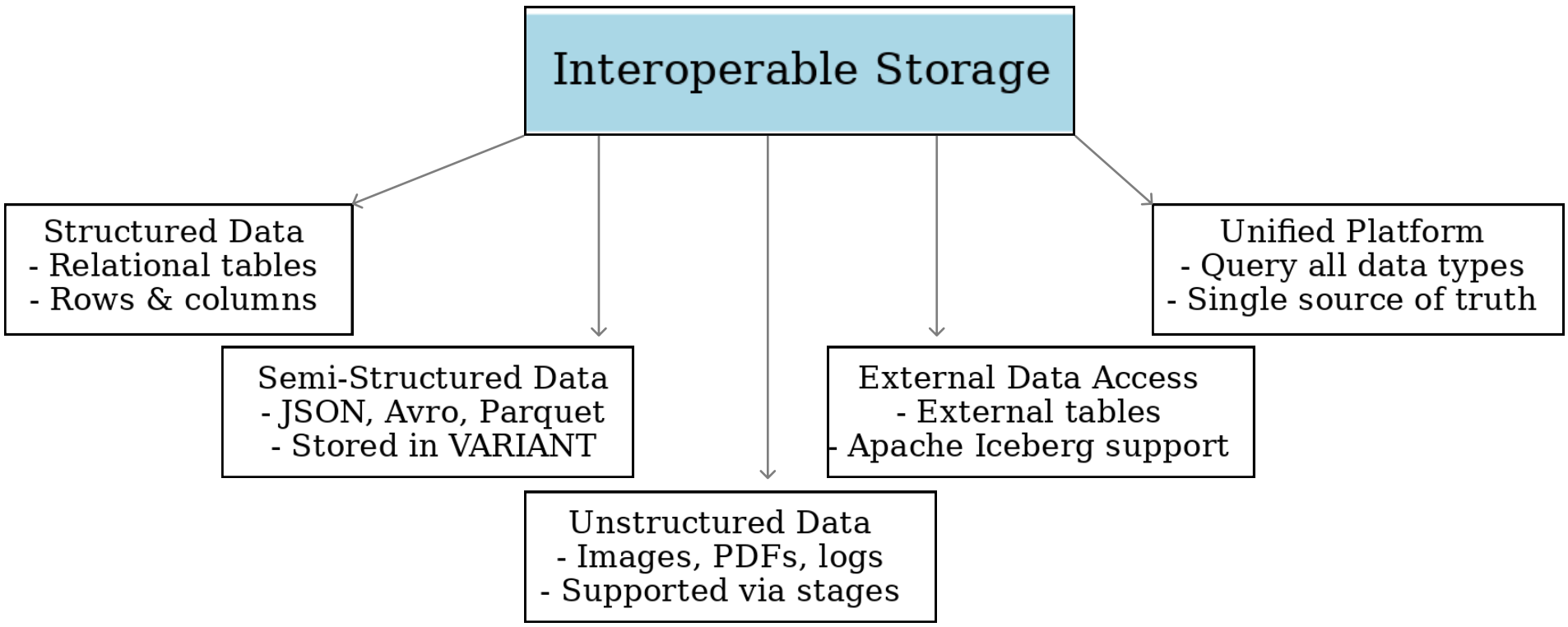
Snowflake's AI Data Cloud is a unified platform that empowers organizations to store, process, and analyze data at scale, with built-in support for advanced analytics and AI/ML. It builds on Snowflake's core cloud data warehouse architecture.

- Interoperable Storage.
- Elastic Compute.
- Snowflake's Layers.
- Snowflake Edition



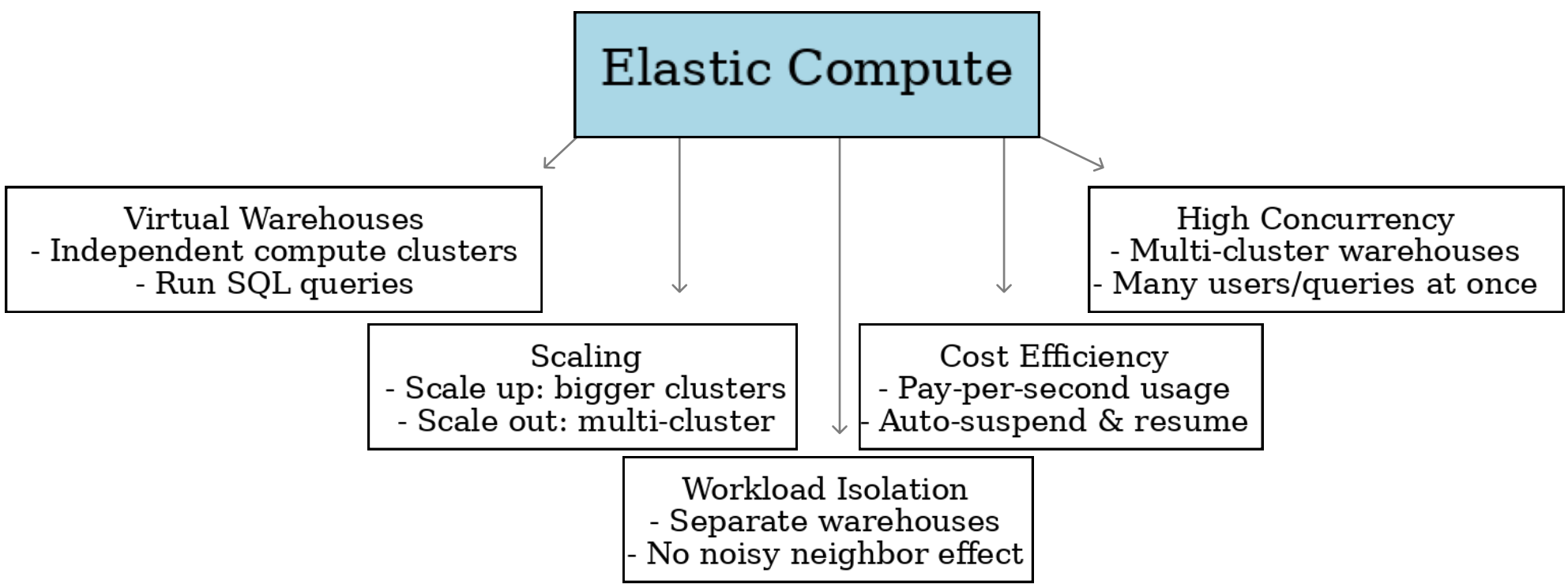
1. Interoperable Storage

Data can reside inside Snowflake's managed storage or externally (e.g. in cloud storage via external tables or open formats like Iceberg) while still being queryable. Snowflake's storage layer automatically organizes data into micro-partitions and compresses it, enabling efficient querying and pruning of large datasets.



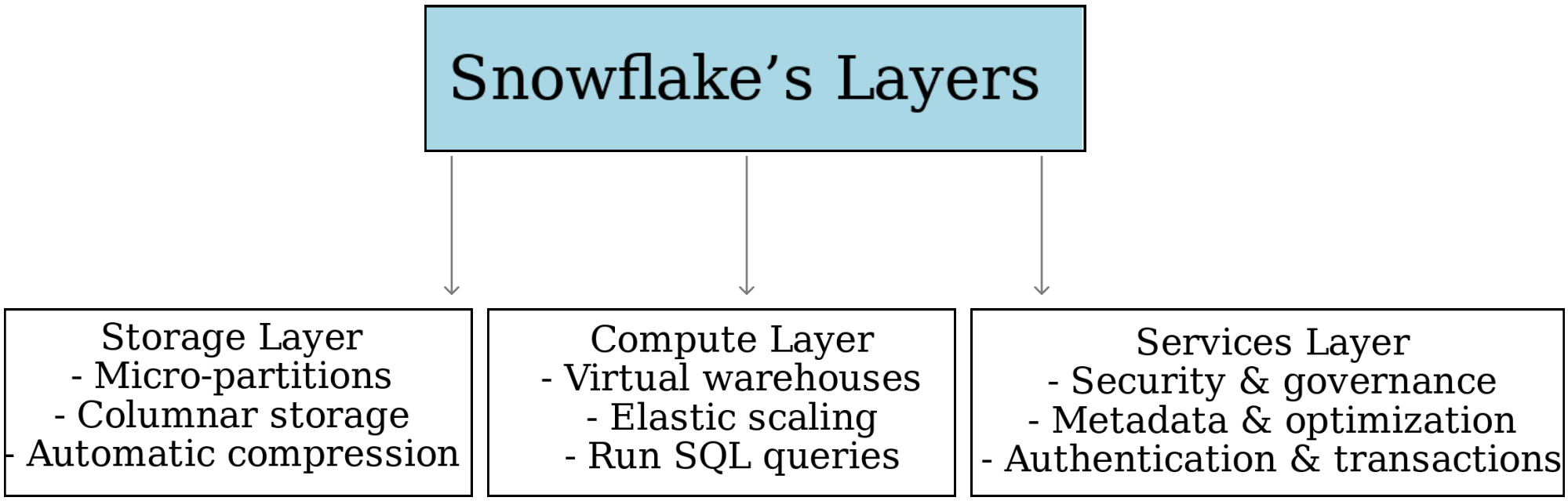
2. Elastic Compute

Snowflake provides isolated compute clusters called virtual warehouses that can scale up or down on demand to handle any workload size or concurrency. Multiple warehouses can run in parallel on the same data without other resource. This multi-cluster, MPP compute design delivers high performance for any number of users or queries by simply adding or resizing compute clusters.



3. Snowflake’s Layers (Architecture)

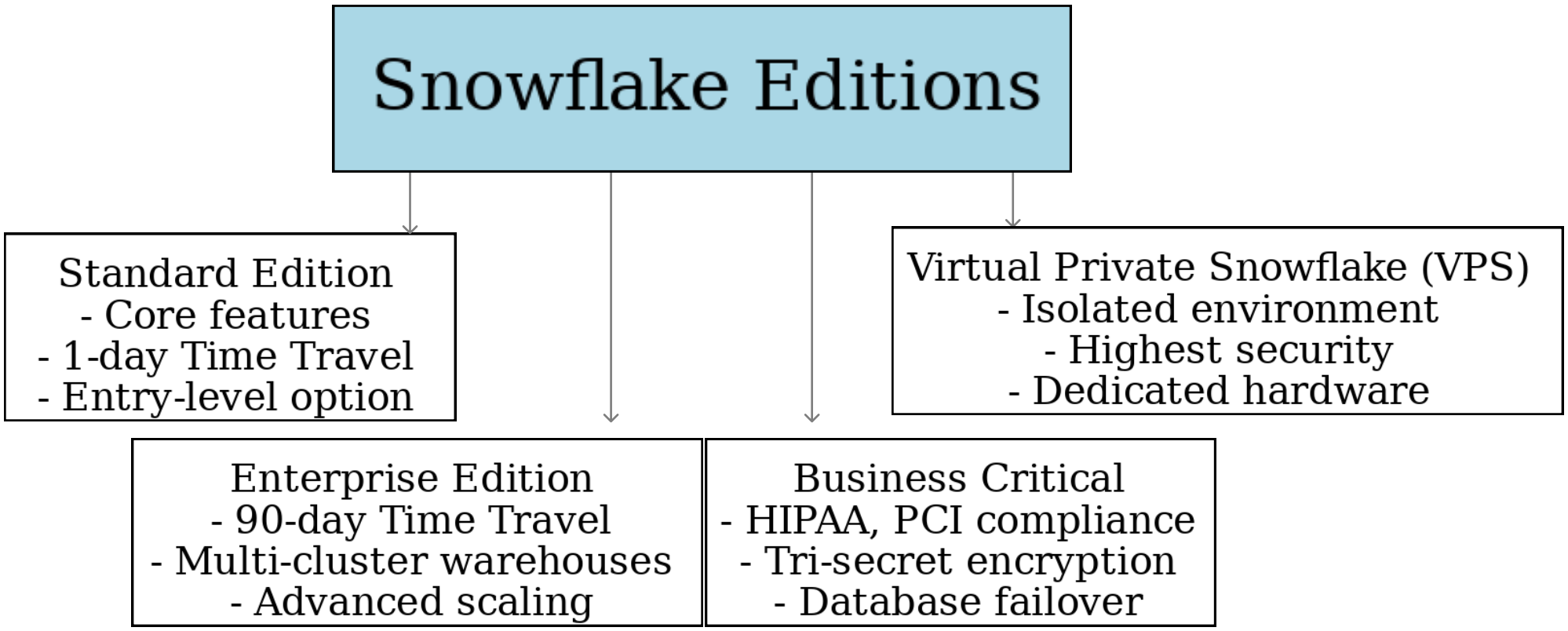
Snowflake’s architecture is logically divided into three main layers. Database Storage, Query Processing (Compute), and Cloud Services.
This layered design a hybrid of shared-disk and shared-nothing that is what enables Snowflake’s elastic scaling and fully managed experience.



4. Snowflake Editions

Snowflake is offered in multiple service editions to suit different needs. Each higher edition builds on the previous, adding features and stronger guarantees as required.

- The Standard Edition.
- The Enterprise Edition.
- Business Critical Edition.
- Virtual Private Snowflake (VPS).



Thank you!

For Questions, Please Comment!

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